

dut.v

File location: \$VCS_HOME/doc/examples/nativetestbench/openvera/rvm/host_if/hdl/dut.v

```
`timescale 1ns/1ns
/*****
*   Top-level module for the memory-mapped register in a dut
*****/
module dut(
    //
    // Intel-style parallel management interface
    //
    Addr, Sel, Data, Rd, Wr, Rdy,

    //
    // Miscellaneous control interfaces
    //
    rst, clk);
    input [4:0] Addr;
    input      Sel;
    inout [7:0] Data;
    input      Rd;
    input      Wr;
    output     Rdy;
    input      rst;
    input      clk;

//
// Register file
//
    reg [7:0]  regfile ['h00:'h1f];
//
// Hardware reset
//
always
begin: do_reset
    integer i;

    wait (rst === 1'b0);
    // Reset the register file
    for (i = 'h00; i <= 'h1f; i = i + 1) begin
```

```

        regfile[i] = 8'h00;
    end
    // Disable all blocks
    disable mgmt_if;
    wait (rst !== 1'b0);
end

//
// Management interface
//

// AC timing characteristics
parameter t4 = 10;
parameter t6 = 05;
parameter t7 = 10;
parameter t9 = 15;
parameter delay = 20;

// Output drivers
reg [7:0] Data_out;
assign    Data = Data_out;
reg       Rdy;

always
begin: mgmt_if
    Data_out = 8'hzz;
    Rdy      = 1'b0;

    // Wait for reset to go away
    wait (rst === 1'b1);
    // Normal operation
    forever begin: do_cycle

        @ (negedge Rd or negedge Wr);
        #(t6);
        Rdy <= 1'b1;
        #(delay);
        Rdy <= 1'b0;
        if (Wr === 1'b0) begin
            // Write cycle
            @ (posedge Wr);
            // Addresses 0-7 is read-only
            if (Addr > 5'h07) regfile[Addr] = Data;
        end
        else begin

```

```
        // Read cycle
        Data_out <= #(t4) regfile[Addr];
        @ (posedge Rd);
        Data_out <= #(t9) 8'hzz;
    end
    Rdy <= #(t7) 1'b0;
end
end
endmodule
```